CS223 Computer Architecture & Organization

Superscalar Processors

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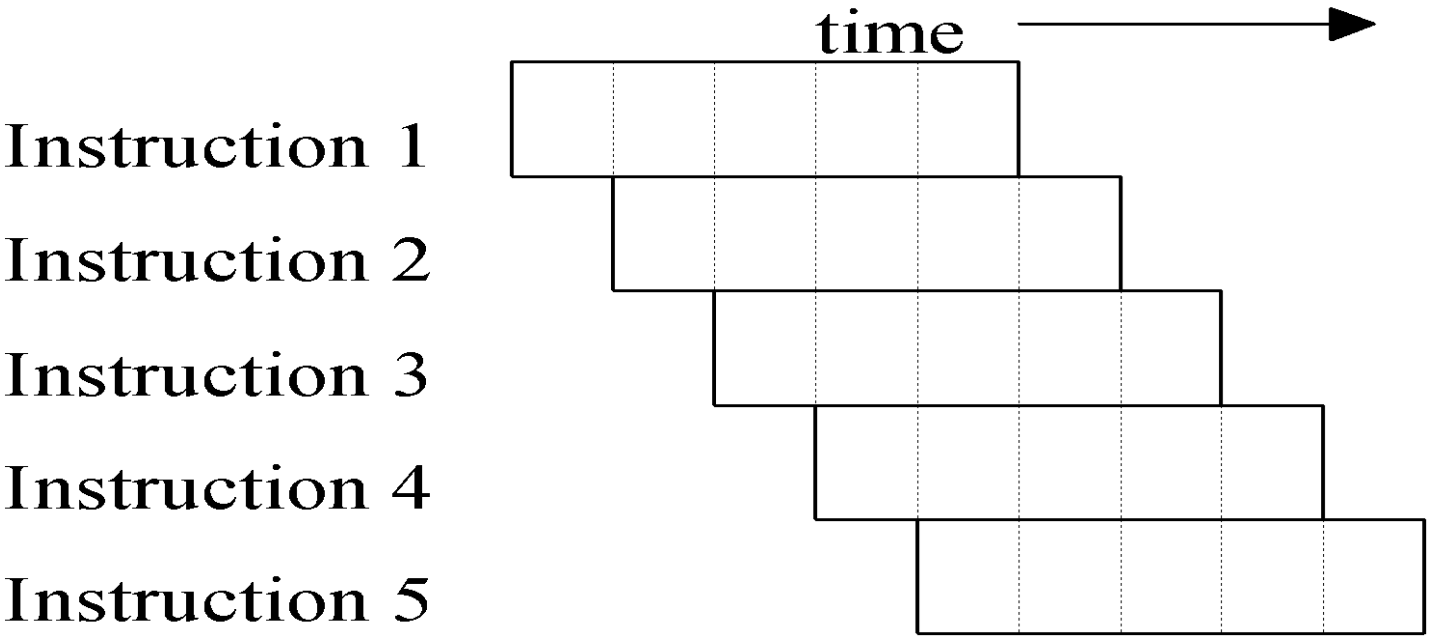
Simple in-order pipelining

Pipelining –goal was to complete one instruction per clock cycle

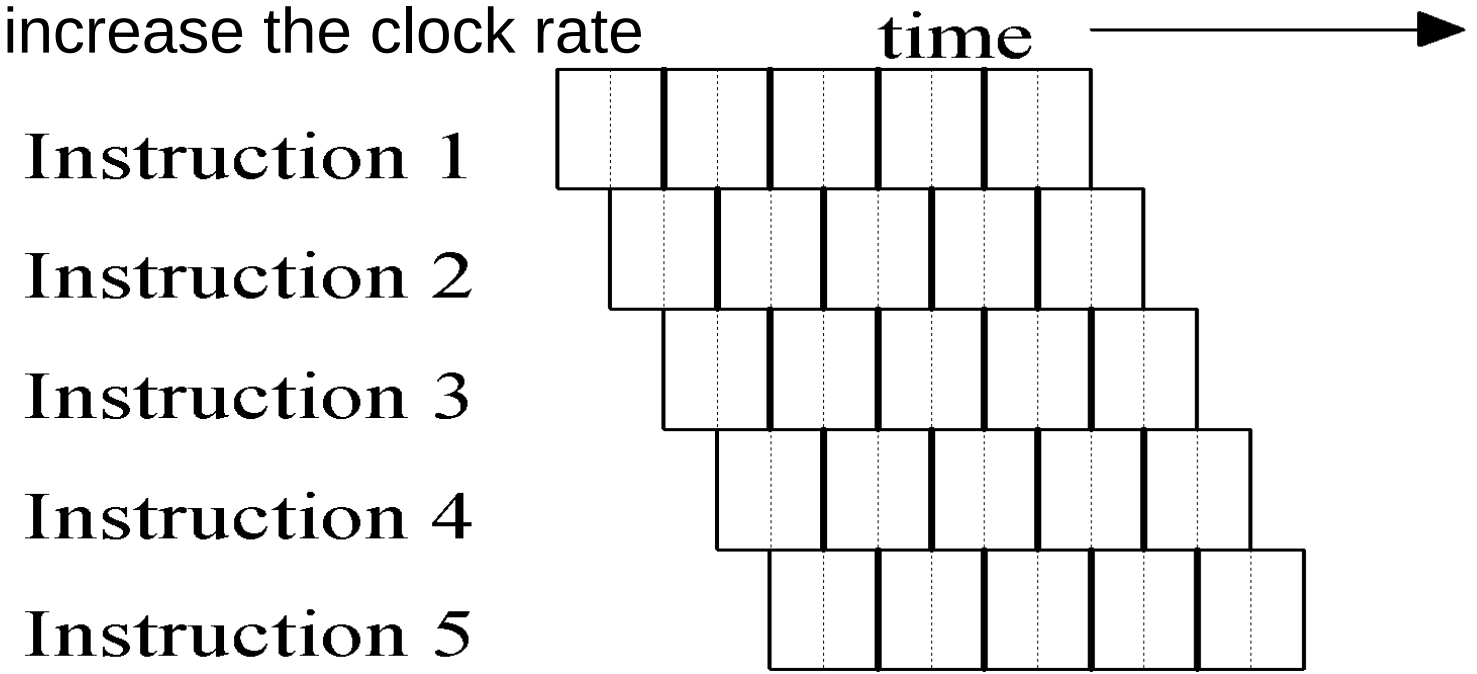


Advanced Pipelining

- Increase the depth of the pipeline to increase the clock rate – **superpipelining**
- Fetch (and execute) more than one instructions at one time (expand every pipeline stage to accommodate multiple instructions) – **multiple-issue (super scalar)**
- Launching multiple instructions per stage allows the instruction execution rate, CPI, to be less than 1

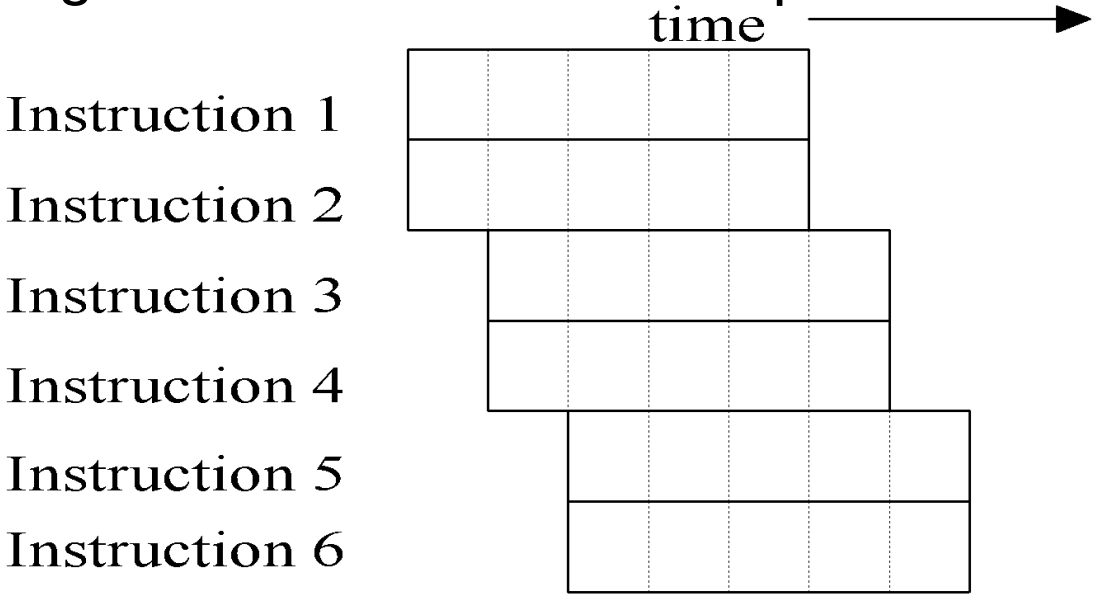
Superpipelining

- superpipelining -Increase the depth of the pipeline to increase the clock rate



Superscalar – multiple-issue

- Fetch (and execute) more than 1 instructions at one time
- Expand every stage to accommodate multiple instructions



Multiple Issue and Static Scheduling

- Basic pipeline will give only a min CPI of 1
- To achieve CPI < 1, need to complete multiple instructions per clock **(superscalar processors)**
- Multiple functional units and Multiple Issue
- Solutions:
  - Statically scheduled superscalar processors -VLIW

    VLIW (very long instruction word) processors
  - Dynamically scheduled superscalar processors

VLIW Processors

- Package multiple operations into one instruction (long word)
- The instruction have a set of fields for each functional unit.
- 16 to 24 bits per unit, → instruction length of 80 to 120 bits.
- Must be enough parallelism in code to fill the available slots
- Find enough parallel instructions by loop unrolling and scheduling

VLIW Processors

- Example VLIW processor:
  - One integer instruction (or branch)
  - Two independent floating-point operations
  - Two independent memory references

			Clock cycle issued
Loop:	L.D	F0,0(R1)	1
	stall		2
	ADD.D	F4,F0,F2	3
	stall		4
	stall		5
	S.D	F4,0(R1)	6
	DADDUI	R1,R1,#-8	7
	stall		8
	BNE	R1,R2,Loop	9

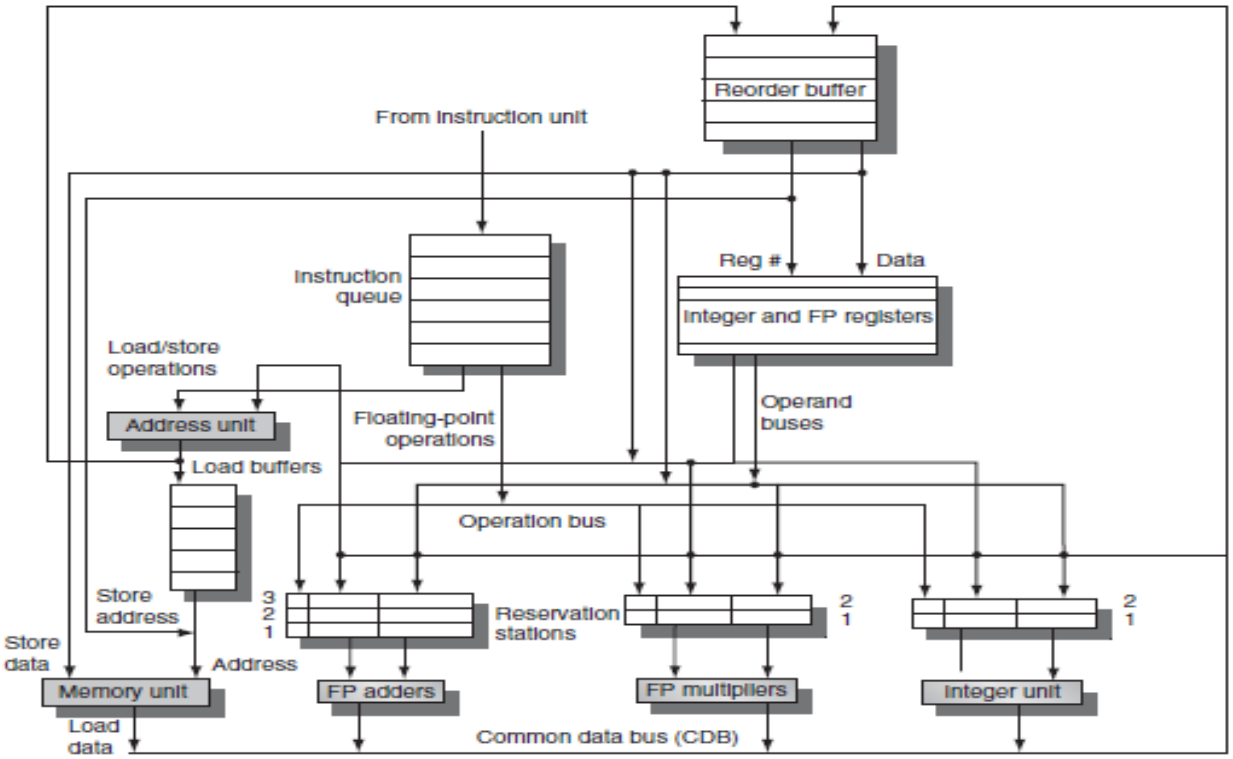
Memory reference 1	Memory reference 2	FP operation 1	FP operation 2	Integer operation/branch
L.D F0,0(R1)	L.D F6,-8(R1)			
L.D F10,-16(R1)	L.D F14,-24(R1)			
L.D F18,-32(R1)	L.D F22,-40(R1)	ADD.D F4,F0,F2	ADD.D F8,F6,F2	
L.D F26,-48(R1)		ADD.D F12,F10,F2	ADD.D F16,F14,F2	
		ADD.D F20,F18,F2	ADD.D F24,F22,F2	
S.D F4,0(R1)	S.D F8,-8(R1)	ADD.D F28,F26,F2		
S.D F12,-16(R1)	S.D F16,-24(R1)			DADDUI R1,R1,#-56
S.D F20,24(R1)	S.D F24,16(R1)			
S.D F28,8(R1)				BNE R1,R2,Loop

- Performance: 9 cycles to complete 26 instructions. More registers used.

Extreme Optimization

- Modern micro-architectures uses this triple combination:
  - Dynamic scheduling + multiple issue + speculation
- Dependency between instructions issued in same clock.
- Register read for multiple instructions in parallel.
- Complex Issue logic to check dependencies and hazards.
- Assign reservation stations and ROB entries in a cycle.

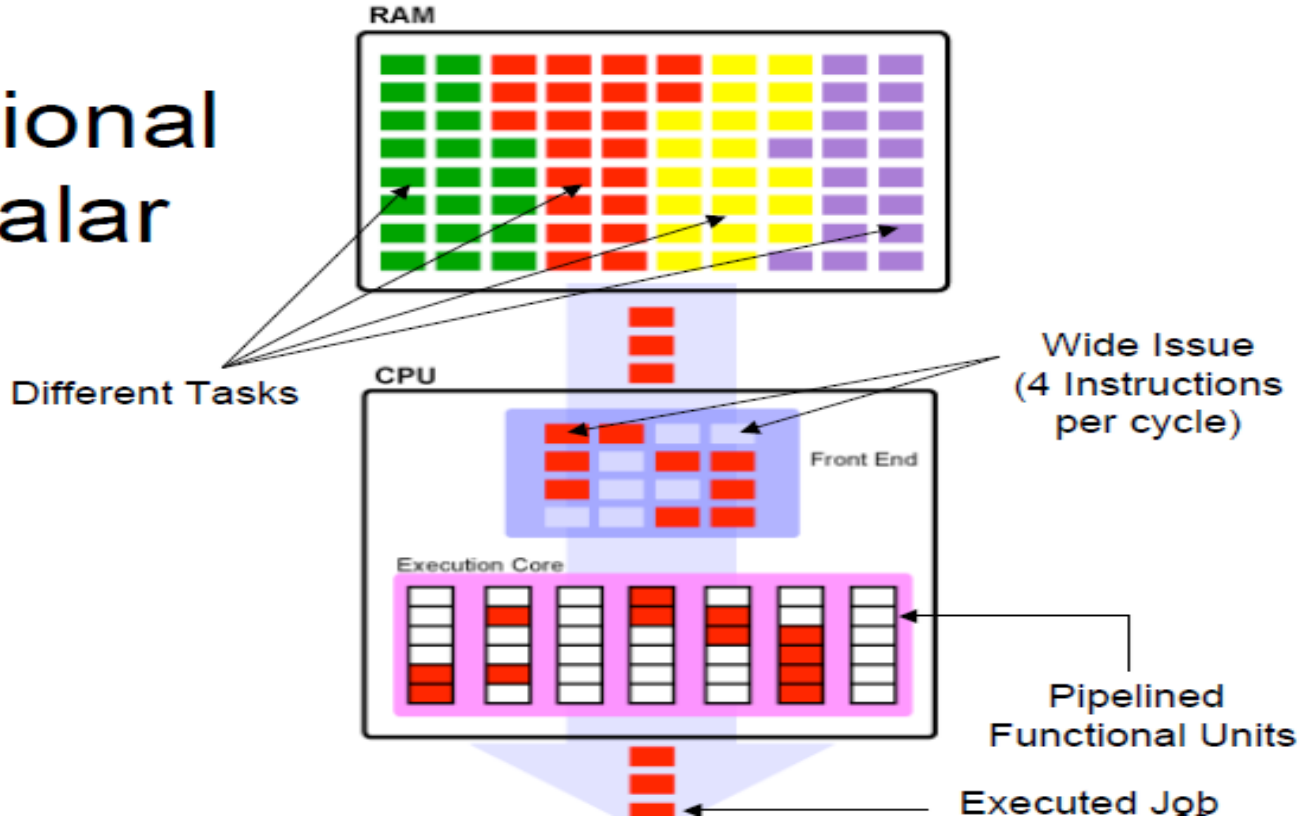
Handling Multiple Issue



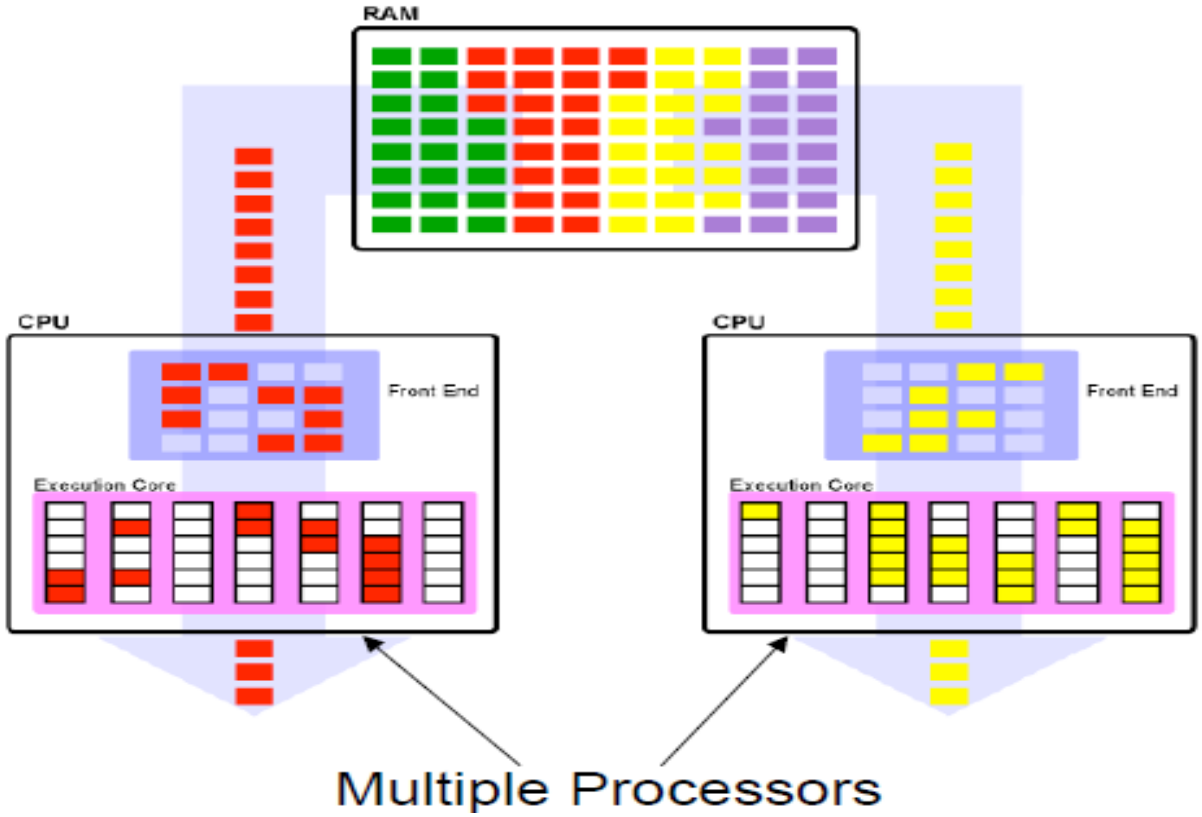
- One issue per unit per cycle
- Wider operand buses, CDB
- Multiple register read/cycle
- Multiple instruction to commit / cycle

Superscalar Processor

Conventional
Superscalar

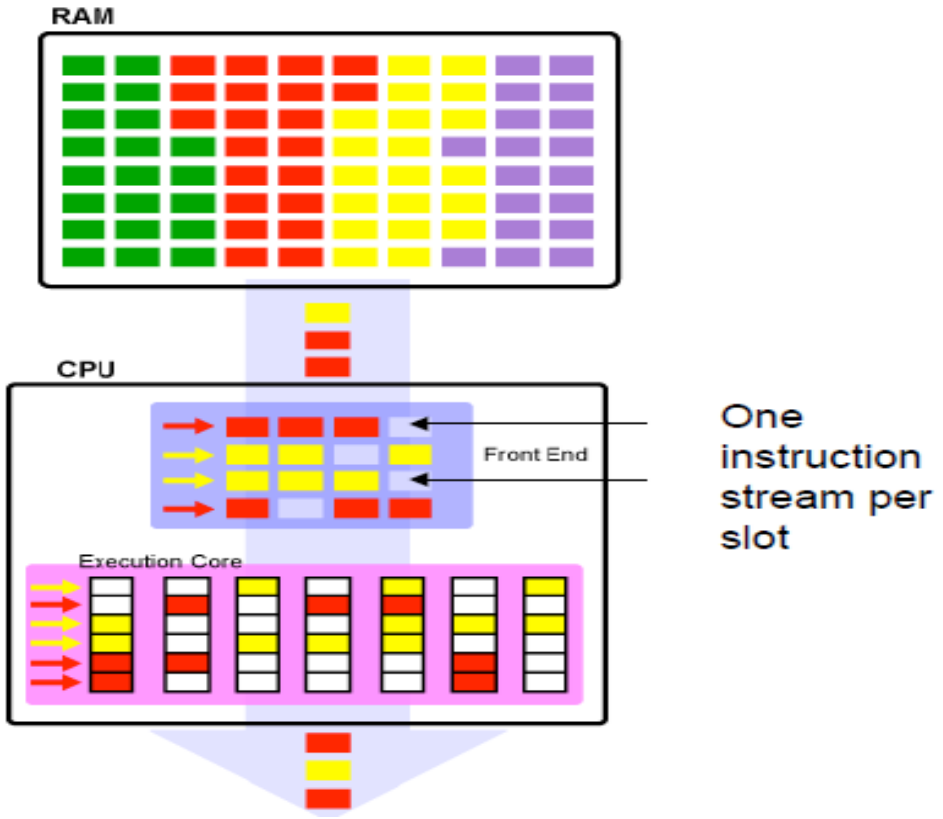


Symmetric Multiprocessor



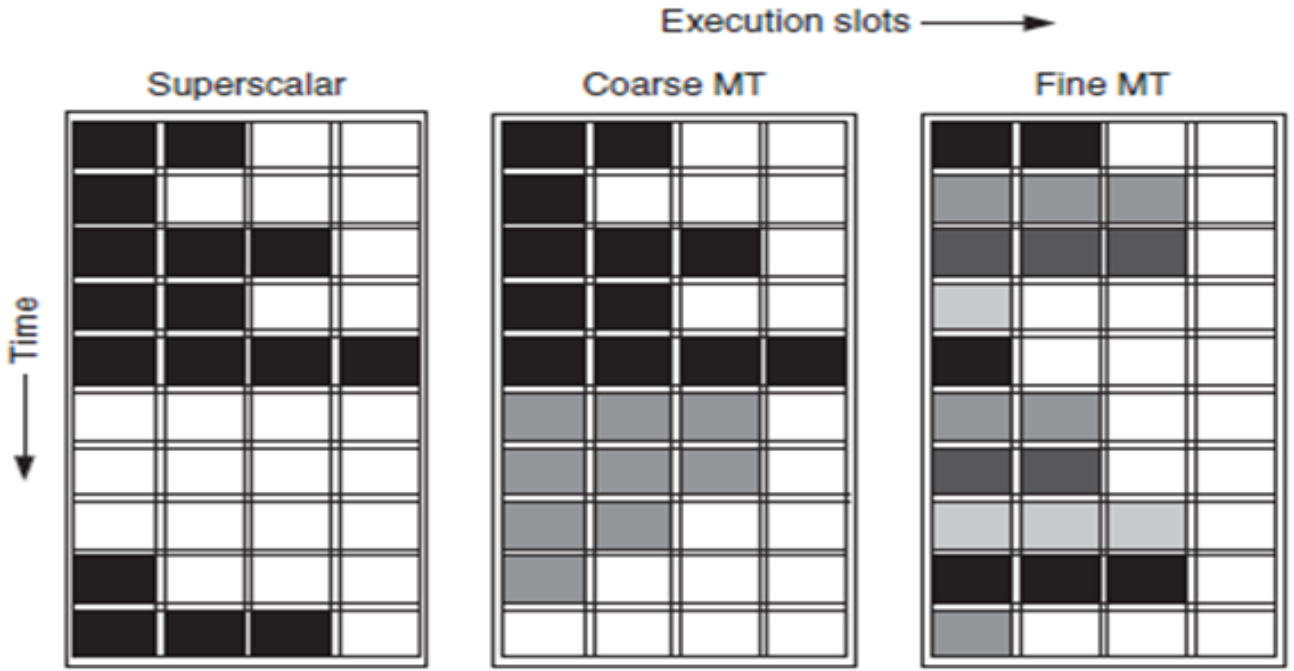
Multithreading

Multithreading



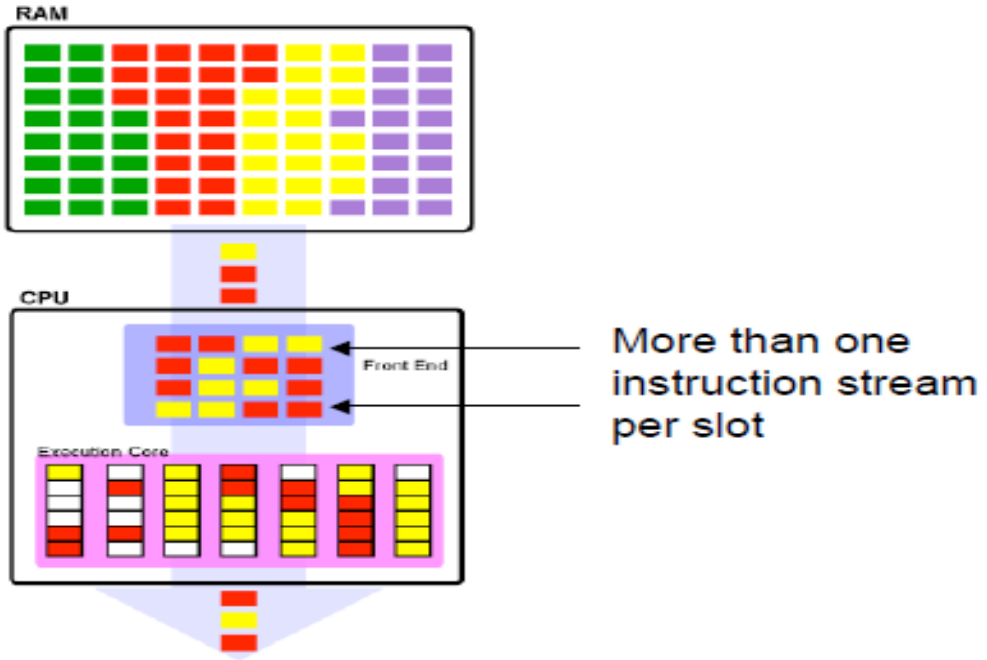
Multithreading

Multithreading allows multiple threads to share the functional units of a single processor in an overlapping fashion

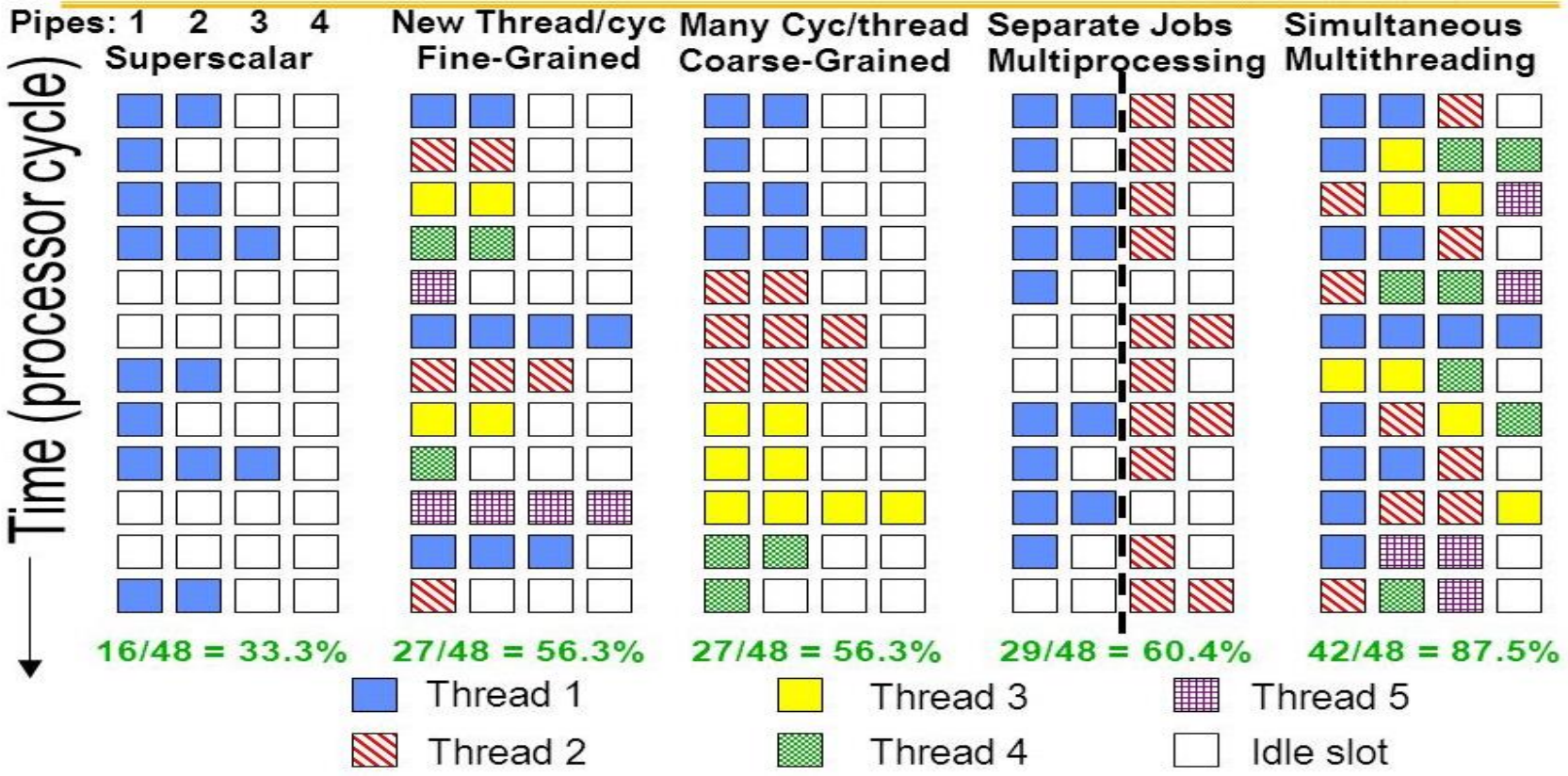


Hyperthreading /SMT

Hyperthreading



Comparison of Resource Utilization



Cache optimization for superscalar

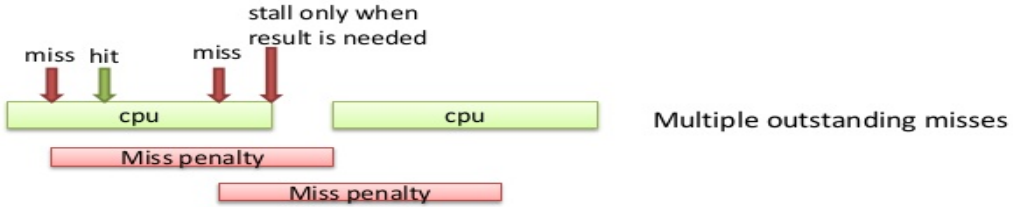
Average memory access time = Hit time + (Miss rate×Miss penalty)

Pipelining Cache

- **Pipelined cache for faster clock cycle time.**
- Split cache memory access into several sub-stages
- Ex: Indexing, Tag Read, Hit/Miss Check, Data Transfer
- Pipeline cache access to improve bandwidth
  - Examples: Pentium 4 – Core i7: 4 cycles
- But slow in hits.
- Increases branch mis-prediction penalty
- Makes it easier to increase associativity

Non-blocking Caches

- **Non blocking caches to increase cache bandwidth**
- Caches can serve hits under multiple cache miss in progress
  - *(a) Hit under miss (b) Hit under multiple miss*



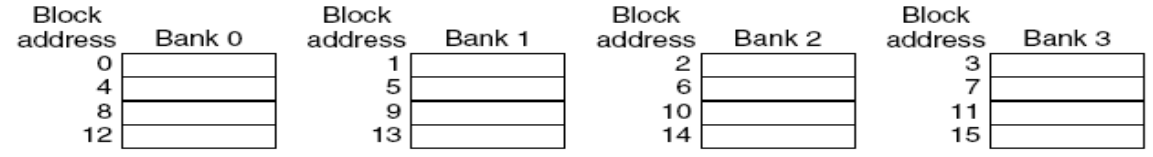
- Must needed for OOO superscalar processor for IPC increase
- L2 must support it with L1-MSHR (Miss Status Holding Reg.)
- On an L1 miss allocate MSHR entry, clear upon L2 respond with cache block reply.

Non-blocking Caches

- **Non blocking caches to increase cache bandwidth**
- Processors can hide L1 miss penalty but not L2 miss penalty
- Reduces the effective miss penalty by overlapping miss latencies
- Significantly increases the complexity of the cache controller as there can be multiple outstanding memory accesses
- Requires pipelined or banked memory system

Multi-banked Caches

- **Multi-banked caches to increase cache bandwidth**
- Rather having a single monolithic block, divide cache into independent banks that can support simultaneous accesses.
  - ARM Cortex-A8 supports 1-4 banks for L2
  - Intel i7 supports 4 banks for L1 and 8 banks for L2
- Sequential Interleaving



Four-way interleaved cache banks using block addressing. Assuming 64 bytes per blocks, each of these addresses would be multiplied by 64 to get byte addressing.

Early Restart

- **Early restart to reduce miss penalty**
- Do not wait for entire block to be loaded for restarting CPU
- **Early restart**
  - Request words in normal order
  - Send missed work to the processor as soon as it arrives
  - L2 controller is not involved in this technique

- **Early restart**

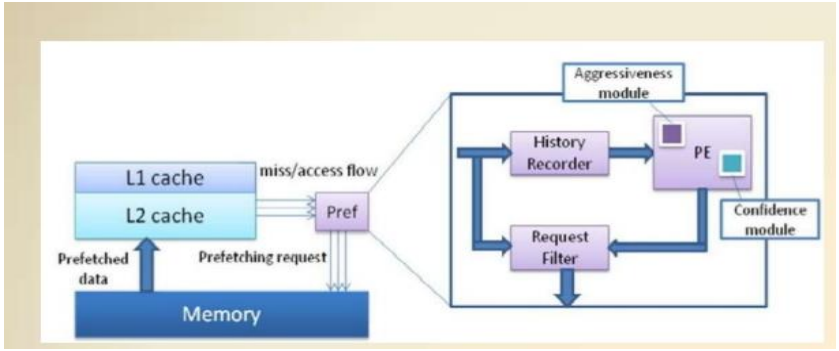
Critical Word First

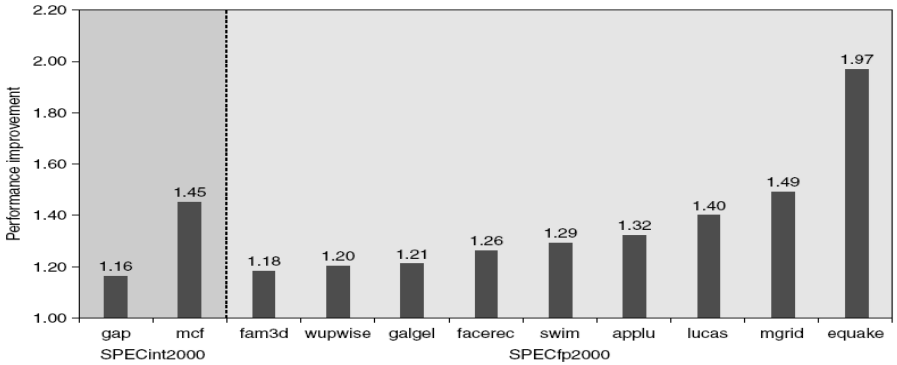
- **Critical word first to reduce miss penalty**
- **Critical word first**
  - Request missed word from memory first
  - Send it to the processor as soon as it arrives
  - Processor resume while rest of the block is filled in cache
  - L2 cache controller forward words of a block out of order.
  - L1 cache controller should re-arrange words in block
    - **Critical word first**

As soon as 3 is brought it is forwarded to CPU
In background 4, 1 and 2 is brought

Hardware Prefetching

- **Pre-fetching to reduce miss rate and miss penalty.**
- Pre-fetch items before processor request them.
- Fetch more blocks on miss -include next sequential block
- Requested block is kept in I-cache and next in stream buffer.
- If a missed block is in stream buffer, cache miss is cancelled





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